

INDIAN INSTITUTE OF TECHNOLOGY MADRAS

Department of Electrical Engineering

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1 Signal Representation & Transmitter Architecture

Based on EE4140 Online Class 1

1.1 Time Domain Description of Digital Bandpass Signals

Digital bandpass signaling relies on digital-to-analog converters (DAC) with oversampling to ensure smooth waveform reconstruction.

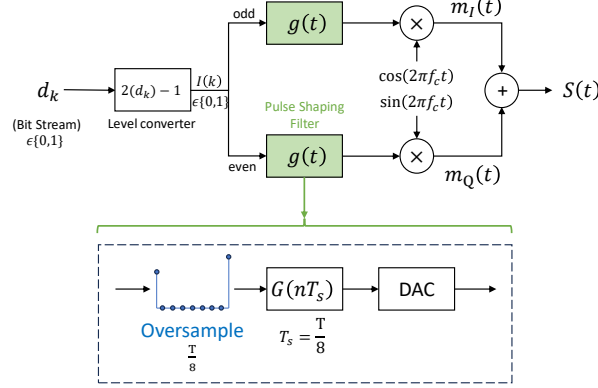


Figure 1: Transmitter Block Diagram.

1.2 In-Phase and Quadrature (I/Q) Modulation

Baseband signals are shaped using pulse filters $g(t)$ (e.g., Square Root Raised Cosine with 88 coefficients):

$$m_I(t) = \sum_k I_I(k)g(t - kT)$$

$$m_Q(t) = \sum_k I_Q(k)g(t - kT)$$

The modulated bandpass signal $s(t)$ is expressed as:

$$S(t) = m_I(t) \cos(2\pi f_c t) - m_Q(t) \sin(2\pi f_c t)$$

Complex Envelope Notation:

$$S(t) = \text{Re} \left\{ \tilde{s}(t) e^{j2\pi f_c t} \right\},$$

where $\tilde{s}(t)$ is the complex baseband signal (complex envelope), given by

$$\tilde{s}(t) = m_I(t) + jm_Q(t).$$

2 Digital Modulation Schemes

1. **PAM** (Pulse Amplitude Modulation)
2. **PSK** (Phase Shift Keying)
3. **QAM** (Quadrature Amplitude Modulation)
4. **FSK** (Frequency Shift Keying)

2.1 Pulse Amplitude Modulation (PAM)

For a k -th symbol duration $kT \leq t < (k+1)T$, the waveform is given by:

$$s_m(t) = A_m g(t) \cos(2\pi f_c t), \quad m = 1, 2, \dots, M$$

where amplitude levels are defined as:

$$A_m = (2m - 1 - M)d$$

For 4-PAM ($M = 4$), $A_m \in \{-3d, -d, d, 3d\}$:

Bit Pattern	Amplitude Level (A_m)
00	$-3d$
01	$-d$
11	$+d$
10	$+3d$

Table 1: 4-PAM Constellation Mapping with Gray Coding.

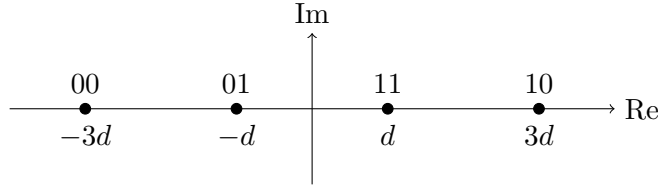


Figure 2: 1D Constellation Layout for 4-PAM.

2.2 Continuous Phase Modulation (CPM)

CPM schemes (CPFSK, MSK) maintain a constant signal envelope. Unlike linear modulation schemes which suffer from envelope fluctuations due to instantaneous phase shifts, CPM prevents power amplifiers being driven out of their linear regions of operation.

3 Probability Analysis for M -ary Orthogonal Signals

Based on EE4140 Online Class 2

For M -ary orthogonal signals with $N = M$ dimensions:

$$s_i(t) = \sqrt{E_s} \phi_i(t), \quad i = 1, 2, \dots, M$$

Assume signal S_1 was transmitted. The received vector $\vec{r} = [r_1, r_2, \dots, r_N]^T$ is:

$$\vec{r} = \begin{bmatrix} \sqrt{E_s} + n_1 \\ n_2 \\ \vdots \\ n_N \end{bmatrix}$$

where $n_j \sim \mathcal{N}(0, \sigma^2)$ with noise variance $\sigma^2 = \frac{N_0}{2}$.

A correct decision is made if $\vec{r}$ falls into the decision region Z_1 , which corresponds to $r_1 > r_2, r_1 > r_3, \dots, r_1 > r_N$:

$$P(C|S_1) = P[\vec{r} \in Z_1|S_1] = P[r_2 < r_1, r_3 < r_1, \dots, r_N < r_1|S_1]$$

Conditioning on the specific value of the first coordinate $r_1 = y$:

$$P(C|S_1) = \int_{-\infty}^{\infty} P[\vec{r} \in Z_1 | S_1, r_1 = y] \cdot p(r_1 = y | S_1) dy$$

Expanding the conditional probability inside the integral:

$$P[\vec{r} \in Z_1 | S_1, r_1 = y] = P[r_2 < y, r_3 < y, \dots, r_N < y | S_1, r_1 = y]$$

Since noise components $n_1, n_2, \dots, n_N$ are mutually independent:

$$P[r_2 < y, r_3 < y, \dots, r_N < y | S_1, r_1 = y] = \prod_{j=2}^N P[r_j < y | S_1]$$

For $j \geq 2$, $r_j \sim \mathcal{N}(0, \frac{N_0}{2})$. The probability $P[r_j < y | S_1]$ is:

$$P[r_j < y | S_1] = \int_{-\infty}^y f_N(x) dx = 1 - \int_y^{\infty} f_N(x) dx$$

where $f_N(x) = \frac{1}{\sqrt{\pi N_0}} \exp\left(-\frac{x^2}{N_0}\right)$.

Since all $N - 1$ un-transmitted dimensions are identically distributed:

$$\prod_{j=2}^N P[r_j < y | S_1] = \left[\int_{-\infty}^y f_N(x) dx \right]^{N-1} = \left[1 - \int_y^{\infty} f_N(x) dx \right]^{M-1}$$

For the transmitted dimension $r_1 = \sqrt{E_s} + n_1$, its probability density function is:

$$p(r_1 = y | S_1) = f_N(y - \sqrt{E_s})$$

Substituting both expressions back into the integral for $P(C|S_1)$:

$$\begin{aligned} P(C|S_1) &= \int_{-\infty}^{\infty} f_N(y - \sqrt{E_s}) \left[\int_{-\infty}^y f_N(x) dx \right]^{M-1} dy \\ &= \int_{-\infty}^{\infty} f_N(y - \sqrt{E_s}) \left[1 - \int_y^{\infty} f_N(x) dx \right]^{M-1} dy \end{aligned}$$

The overall probability of error $P(e)$ is:

$$P(e) = 1 - P(C) = 1 - \int_{-\infty}^{\infty} f_N(y - \sqrt{E_s}) \left[1 - \int_y^{\infty} f_N(x) dx \right]^{M-1} dy$$

4 Probability Analysis for Bi-Orthogonal Signals

For $M = 2N$ bi-orthogonal signals constructed using N orthogonal basis vectors:

$$s_1(t) = \sqrt{\frac{2E_s}{T}} \cos(2\pi f_c t), \quad s_2(t) = -s_1(t)$$

Assume S_1 was transmitted. The received vector is $\vec{r} = [\sqrt{E_s} + n_1, n_2, \dots, n_N]^T$. A correct decision requires $r_1 > 0$ and $|r_j| < r_1$ for all $j = 2, 3, \dots, N$:

$$P(C|S_1) = \int_0^{\infty} P[|r_2| < y, |r_3| < y, \dots, |r_N| < y | S_1, r_1 = y] p(r_1 = y | S_1) dy$$

Due to independence across dimensions:

$$P[|r_j| < y | S_1] = P[-y < r_j < y | S_1] = \int_{-y}^y f_N(x) dx$$

Expressing this in terms of the upper tail integral:

$$\int_{-y}^y f_N(x) dx = 1 - 2 \int_y^\infty f_N(x) dx$$

Taking the product across all $N - 1$ non-signal dimensions:

$$\prod_{j=2}^N P[|r_j| < y | S_1] = \left[1 - 2 \int_y^\infty f_N(x) dx \right]^{N-1}$$

Substituting into the outer integration over $y \in (0, \infty)$:

$$P(C|S_1) = \int_0^\infty f_N(y - \sqrt{E_s}) \left[1 - 2 \int_y^\infty f_N(x) dx \right]^{N-1} dy$$

5 Union Bound Derivations

5.1 Standard Union Bound

Let $\mathcal{E}_{1k}$ be the pairwise error event that S_1 is wrongly decoded as S_k :

$$P(\vec{r} \notin Z_1 | S_1) = P(\mathcal{E}_{12} \cup \mathcal{E}_{13} \cup \dots \cup \mathcal{E}_{1M})$$

Applying Boole's inequality (Union Bound):

$$P(A \cup B) \leq P(A) + P(B) \implies P(\vec{r} \notin Z_1 | S_1) \leq \sum_{k=2}^M P(\mathcal{E}_{1k})$$

For BPSK/QPSK signal constellations:

$$P(C) = (1 - q)^2 \implies P(e) = 1 - P(C) = 2q - q^2 < 2q$$

where $q = Q\left(\frac{d_{\min}}{\sqrt{2N_0}}\right)$.

5.2 Intelligent Union Bound (IUB)

By retaining only the $N_{\text{neighbors}}$ nearest neighbor symbols in the constellation:

$$P_{\text{IUB}}(e) = N_{\text{neighbors}} \cdot Q\left(\frac{d_{\min}}{\sqrt{2N_0}}\right)$$

6 Maximum Likelihood Sequence Estimation (MLSE) & Viterbi Algorithm

Based on EE4140 Online Class 4

6.1 System Model

For a discrete-time inter-symbol interference (ISI) channel with memory length $L - 1$:

$$r(k) = \sum_{l=0}^{L-1} f_l I(k-l) + n(k) = f_0 I(k) + f_1 I(k-1) + \dots + f_{L-1} I(k-L+1) + n(k)$$

6.2 Trellis Structure

The channel state at time step k is $\mathbf{S}_k = [I(k-1), I(k-2), \dots, I(k-L+1)]$. Total number of states = 2^{L-1} for binary inputs.

- **Branch Metric:** $M_k = (r(k) - \hat{r}(k))^2$
- **Cumulative Metric:** $CM_k = CM_{k-1} + M_k$
- **Survivor Selection:** At each node, retain incoming path with minimum CM_k .
- **Truncation Depth:** Practical survivor memory length $D \approx 5L$ to $6L$.

7 Equalization Techniques for ISI Channels

Equalizers are broadly categorized into **Linear Equalizers** (Zero Forcing (ZF), Linear MMSE (LMMSE)) or **Non-Linear Equalizers** (Decision Feedback Equalizer (DFE), MLSE (Viterbi))

7.1 Infinite Order Zero Forcing (ZF)

Forces total ISI to zero:

$$F(z)W(z) = 1 \implies W(z) = \frac{1}{F(z)}$$

7.2 Infinite Order LMMSE Equalizer

Minimizes mean squared error $J = \min_w E\{|e(k)|^2\}$:

$$W(z) = \frac{F^*(z^{-1})}{F(z)F^*(z^{-1}) + N_0}$$

Orthogonality Principle: The error is orthogonal to the observation data:

$$E\{e(k)y^*(k-j)\} = 0, \quad \forall j$$

8 Bayesian / Linear Minimum Mean Squares Estimator

Consider a 2-tap wireless channel model:

$$y(k) = h(0)x(k) + h(1)x(k-1) + v(k)$$

Stacking 3 consecutive observations $\vec{y}(k) = [y(k), y(k-1), y(k-2)]^T$:

$$\begin{bmatrix} y(k) \\ y(k-1) \\ y(k-2) \end{bmatrix} = \begin{bmatrix} h(0) & h(1) & 0 & 0 \\ 0 & h(0) & h(1) & 0 \\ 0 & 0 & h(0) & h(1) \end{bmatrix} \begin{bmatrix} x(k) \\ x(k-1) \\ x(k-2) \\ x(k-3) \end{bmatrix} + \begin{bmatrix} v(k) \\ v(k-1) \\ v(k-2) \end{bmatrix}$$

$$\vec{y}(k) = H\vec{x}(k) + \vec{v}(k)$$

We seek a linear estimate $\hat{x} = \vec{h}^T \vec{y}$. From $\vec{y}(k)$, we estimate $x(k)$, thereby eliminating the effect of the interfering symbols $x(k-1)$, $x(k-2)$, and $x(k-3)$.

Objective: Minimize $J(\vec{h}) = E\{(\vec{h}^T \vec{y} - x)^2\}$.

Expanding the squared error term:

$$\begin{aligned}
J(\vec{h}) &= E \left\{ (\vec{h}^T \vec{y} - x)^2 \right\} \\
&= E \left\{ (\vec{h}^T \vec{y} - x)(\vec{h}^T \vec{y} - x)^T \right\} \\
&= E \left\{ (\vec{h}^T \vec{y} - x)(\vec{y}^T \vec{h} - x) \right\} \\
&= E \left\{ \vec{h}^T \vec{y} \vec{y}^T \vec{h} - x \vec{y}^T \vec{h} - \vec{h}^T \vec{y} x + x^2 \right\} \\
&= E \left\{ \vec{h}^T \vec{y} \vec{y}^T \vec{h} - 2 \vec{h}^T E \{ \vec{y} x \} + x^2 \right\} \\
&= \vec{h}^T E \{ \vec{y} \vec{y}^T \} \vec{h} - 2 \vec{h}^T E \{ \vec{y} x \} + E \{ x^2 \} \\
&= \vec{h}^T R_{yy} \vec{h} - 2 \vec{h}^T \vec{r}_{yx} + r_{xx}.
\end{aligned}$$

Differentiating with respect to the filter coefficient vector $\vec{h}$:

$$\frac{dJ(\vec{h})}{d\vec{h}} = 2R_{yy}\vec{h} - 2\vec{r}_{yx} = 0$$

$$2R_{yy}\vec{h} = 2\vec{r}_{yx} \implies \vec{h} = R_{yy}^{-1}\vec{r}_{yx} \text{ (Channel inverse estimates)}$$

Thus, the optimal LMMSE estimate is:

$$\hat{x}_{\text{LMMSE}} = \vec{h}^T \vec{y} = \vec{r}_{yx}^T R_{yy}^{-1} \vec{y}$$

For independent IID symbols with variance P_d and noise variance σ^2 :

$$\begin{aligned}
R_{yy} &= E \{ (H\vec{x} + \vec{v})(H\vec{x} + \vec{v})^T \} \\
&= E \{ H\vec{x}\vec{x}^T H^T + H\vec{x}\vec{v}^T + \vec{v}\vec{x}^T H^T + \vec{v}\vec{v}^T \} \\
&= HE \{ \vec{x}\vec{x}^T \} H^T + E \{ \vec{v}\vec{v}^T \} \quad (\text{since } \vec{x} \text{ and } \vec{v} \text{ are uncorrelated}) \\
&= P_d H H^T + \sigma^2 I.
\end{aligned}$$

Evaluating the cross-correlation vector $\vec{r}_{xy} = E \{ x(k) \vec{y}^T(k) \}$:

$$\begin{aligned}
\vec{r}_{xy} &= E \{ x(k) (H\vec{x}(k) + \vec{v}(k))^T \} \\
&= E \{ x(k) \vec{x}^T(k) \} H^T + 0 \\
&= E \{ x(k) [x(k) \quad x(k-1) \quad x(k-2) \quad x(k-3)] \} H^T \\
&= [P_d \quad 0 \quad 0 \quad 0] H^T = P_d \mathbf{e}_1^T H^T.
\end{aligned}$$

Substituting R_{yy} and $\vec{r}_{xy}$ into the LMMSE estimator expression yields:

$$\hat{x}(k) = P_d \mathbf{e}_1^T H^T (P_d H H^T + \sigma^2 I)^{-1} \vec{y}(k).$$

9 Decision Feedback Equalizer (DFE)

Based on EE4140 Online Class 5 The DFE comprises a fractionally spaced feed-forward filter (FFF), a symbol-spaced feedback filter (FBF), a decision device, and an error computation block. The received signal $r(k)$ is processed by the $T/4$ -spaced (T is symbol duration) FFF, which suppresses precursor ISI and compensates for channel distortion. Its output is combined with the feedback contribution from previously detected symbols. The resulting equalized signal $y(k - \delta)$ is passed through the decision device to obtain the hard decision $\hat{a}(k)$.

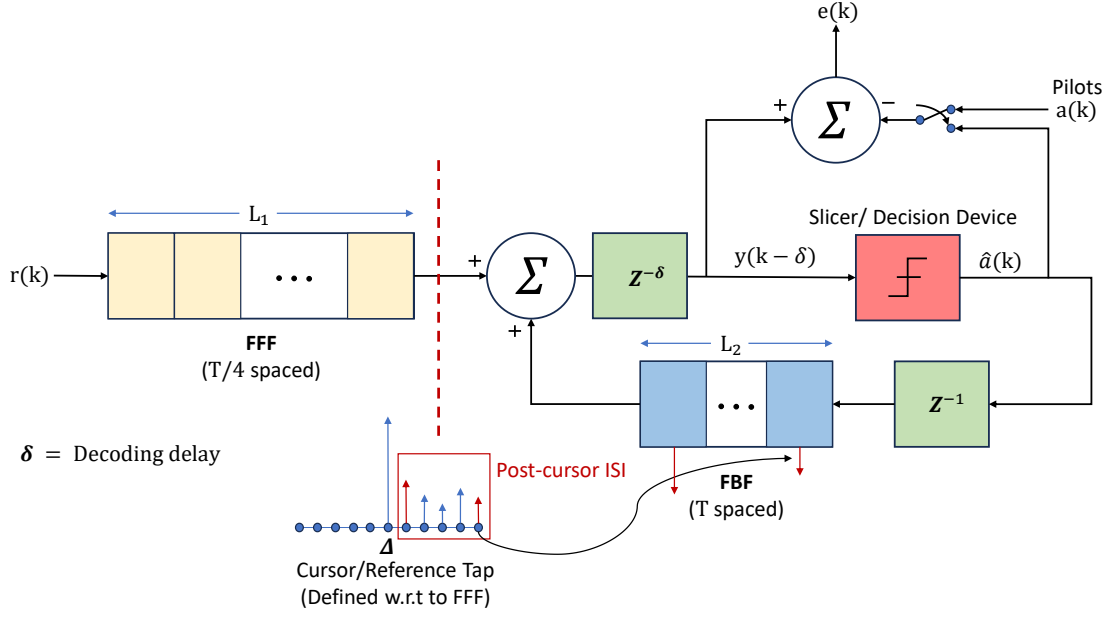


Figure 3: Decision Feedback Equalizer (DFE) with fractionally spaced FFF and symbol spaced FBF.

The FBF operates at the symbol rate and uses previously detected symbols to estimate and cancel post-cursor ISI. The decoding delay δ aligns the equalizer output with the symbol being detected, while the cursor or reference tap, denoted by Δ , specifies the reference position within the FFF.

For a $T/4$ -spaced FFF, if Δ denotes the cursor position in units of FFF tap intervals, the corresponding decoding delay is $\delta = \Delta T/4$, or equivalently $\delta_{\text{sym}} = \Delta/4$ when expressed in symbol periods. For a symbol-spaced FFF, the tap spacing is T , giving $\delta = \Delta T$ and hence $\delta_{\text{sym}} = \Delta$. During training, the known pilot symbols are used as the reference for computing the error $e(k) = a(k) - y(k - \delta)$, whereas during decision-directed operation, the detected symbols are used, giving $e(k) = \hat{a}(k) - y(k - \delta)$. The resulting error drives the adaptive update of the FFF and FBF coefficients via LMS or NLMS adaptation.

9.1 Why Fractionally Spaced Equalization?

- T -spaced equalizers suffer from aliasing prior to equalization if signal bandwidth exceeds $\frac{1}{2T}$.
- $T/2$ -spaced equalizers sample at twice the symbol rate:

$$Y_{T/2}(f) = C_{T/2}(f)X(f)e^{j2\pi f\tau_0}$$

- Fractionally Spaced Equalization synthesizes optimal matched filtering and equalization prior to aliasing, eliminating sensitivity to receiver sampling delay τ_0 .

9.2 MATLAB Implementation

The following example implements a $T/4$ -spaced DFE with LMS adaptation over a two-path, 3-Hz BiGaussian Rayleigh fading channel. The first `numTrainingSym` symbols are used for training, followed by decision-directed adaptation.

Listing 1: Example MATLAB implementation of the $T/4$ -spaced DFE.

```

M = 4;
numSym = 1800;
numTrainingSym = 200;
numPackets = 5;
SNR = 35;
refTap = 5; % Cursor / reference tap
nFwdTaps = 9; % T/4-spaced FFF
nFdbkTaps = 4; % T-spaced FBF
stepsz = 0.01;
doppBiGaussian = doppler("BiGaussian");

raylChan = comm.RayleighChannel( ...
    'PathDelays', [0 2e-6], ...
    'AveragePathGains', [0 -9], ...
    'MaximumDopplerShift', 3, ...
    'PathGainsOutputPort', true, ...
    'SampleRate', 1e6, ...
    'DopplerSpectrum', doppBiGaussian);

dfeq1 = comm.DecisionFeedbackEqualizer( ...
    'Algorithm', 'LMS', ...
    'NumForwardTaps', nFwdTaps, ...
    'NumFeedbackTaps', nFdbkTaps, ...
    'ReferenceTap', refTap, ...
    'StepSize', stepsz, ...
    'TrainingFlagInputPort', true, ...
    'InputSamplesPerSymbol', 4);

% Tx and Rx raised-cosine filters
filt = comm.RaisedCosineTransmitFilter("RolloffFactor", 0.35, ...
    "OutputSamplesPerSymbol", 4);
filt2 = comm.RaisedCosineReceiveFilter("RolloffFactor", 0.35, ...
    "InputSamplesPerSymbol", 4, "DecimationFactor", 1);

% Combined Tx/Rx filter group delay: 20 + 20 = 40 samples
dfeq1.InputDelay = 40;

for p = 1:numPackets
    trainingFlag = true;

    for q = 1:(numSym + numTrainingSym)

        data = randi([0 M-1]);
        tx = pskmod(data, M, pi/4);
        txup = filt(tx);
        [xc, pg] = raylChan(txup);
        rx = awgn(xc, SNR);
        rxmatch = filt2(rx);

        [y, err, wts] = dfeq1(rxmatch, tx, trainingFlag);

        if q == numTrainingSym
            trainingFlag = false;
        end
    end
end
end

```

The main correspondence with Fig. 3 is as follows: `rxmatch` is the DFE input $r(k)$; `NumForwardTaps` = 9 specifies the $T/4$ -spaced FFF; `ReferenceTap` = 5 defines the cursor/reference tap; `NumFeedbackTaps` = 4 specifies the symbol-spaced FBF; `InputSamplesPerSymbol` = 4 sets the fractionally spaced operation at $T/4$; `InputDelay` = 40 provides alignment for the combined Tx/Rx filter group delay; `y` denotes the equalizer output $y(k - \delta)$; `err` denotes the adaptation error $e(k)$; and `wts` contains the combined adaptive FFF and FBF coefficients. The first 200 symbols are used as known training symbols; thereafter, the equalizer operates in decision-directed mode. The two channel paths have average powers of 0 and -9 dB and delays of 0 and $2 \mu\text{s}$, respectively.

10 Single Carrier Block Modulation vs. Multi-Carrier OFDM

Feature	SC-FDE	OFDM
Modulation Domain	Time Domain	Frequency Domain
Equalization Domain	Frequency Domain (IFFT at Rx)	Frequency Domain (1-Tap at Rx)
PAPR	Low	High
Cyclic Prefix Requirement	$N_{cp} > L$	$N_{cp} > L$

Table 2: Comparison of Single Carrier FDE and Multi-Carrier OFDM.

The Cyclic Prefix (CP) converts linear channel convolution into circular convolution, enabling computationally efficient 1-tap equalization in the frequency domain.